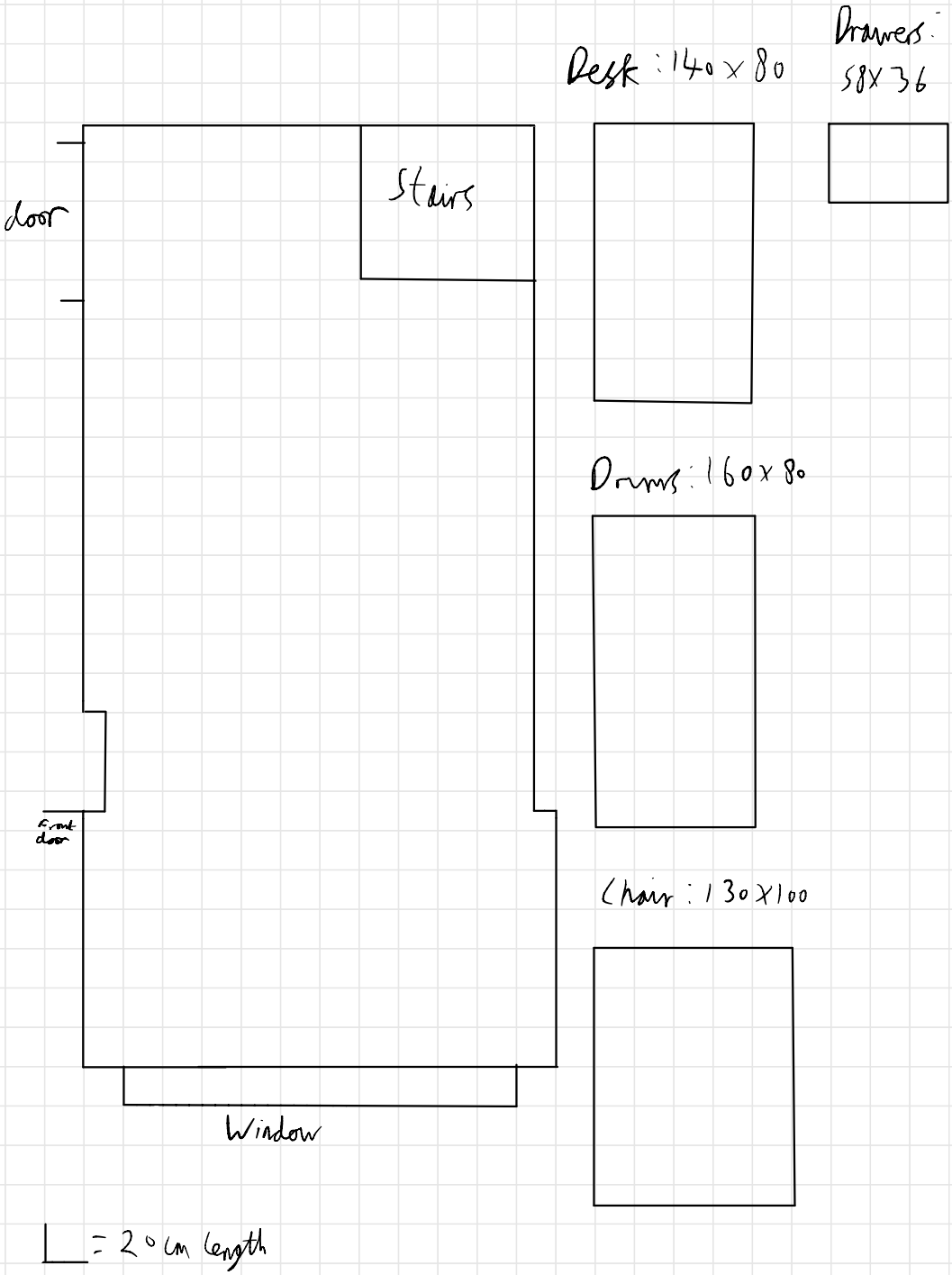
Weekly Plan

	MON	TUE	WED	THU	FRI	SAT	SUN
9.00			Maths		Maths		
						Orchestra	
11.20		Maths	Progress		Chemistry		
		Lunch			Lunch		
1.10		Maths			Chemistry		
2.15						Golf	
2.30		Chemistry	Chess	Maths			
3.30							
4.00			Flute		Orchestra		
4.30	Drumming						
5.30							


Desk : 140 x 80
Drawers
58 x 36
Stairs
door
Drums : 160 x 80
front door
Chair : 130 x 100
Window
= 20 cm length

Notes

Branston baked beans. Farmfoods, Iceland. Waitrose

March 2023

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
		1st	2nd	3rd Small angles my maths ✓	4th	5th
6th	7th Bond skills task	8th	9th Michael Skills task	10th Chem sheet	11th	12th
13th	14th	15th	16th			

	☐
	☐
	☐
	☐
	☐
	☐
	☐
	☐
	☐
	☐
	☐
	☐
	☐
	☐

28/02	☑ Binomial partial fractions my maths
28/02	☑ Radians, arcs and sectors my maths
01/03	☑ Micheal's skills task
27/02	☑ 5 a day
28/02	☑ 5 a day
01/03	☑ 5 a day
28/03	☑ Barbara skills task
02/03	☑ 5 a day
03/03	☑ Barbara Small angle aproximations
03/03	☑ 5 a day
	☐ Pay for cinema ticket
02/03	☑ Transforming My Maths.
09/03	☐ Skills task micheal.
07/03	☐ Barbara Skills task

Mon 27th Feb

13:44 Tired. Slow morning. Done skills task and 5 a day. Breakfast banana, lunch mac and cheese.

15:21 Hallanni bites + Countdown.

17:52 Drumming done. Got two My Maths to do.

20:28 My maths x2 done. Stir fry dinner.

21:56 Teams clear for the week. Chocolate + bed.

Tues 28th Feb

20:43 Normal day at college. Lovely chicken dinner. Lights all at 10 with sam. Strike day tomorrow, Cocaine bear cinema.

Wed 1st March

19:23 Cooking Scampi + chips. Seen cocaine bear. Badminton with reece, will, matt. Strike day.

Thurs 2nd March

17:29 Doing 5 a day. 1 to 1 and maths.

18:13 Going to do My Maths + tomorrow's 5 a day.

19:00 5 a day done. Eating burgers now. My maths after.

Friday 3rd March

27/02

$(1+4x)^4$

$\binom{4}{0}1^4 + \binom{4}{1}1^3 \times 4x + \binom{4}{2}1^2 \times (4x)^2$

$6 \times 16x^2 = 96x^2$

$\frac{4x}{(1-4x)^2} \approx (4x)(1-4x)^{-2}$

$(1-4x)^{-2} = 1 + (-2)(-4x) + \frac{(-2)(-3)}{2}(-4x)^2$

$1 + 8x + 48x^2$

$\times (4x)$

$4x + 32x^2 + 192x^3$

$|4x| < 1$

$|x| < \frac{1}{4}$

$\frac{1}{(1+8x)(1+x)^2} = \frac{A}{1+8x} + \frac{B}{1+x} + \frac{C}{(1+x)^2}$

$\frac{A(1+x)^2}{(1+8x)(1+x)^2} + \frac{B(1+8x)(1+x)}{(1+8x)(1+x)^2} + \frac{C(1+8x)}{(1+8x)(1+x)^2}$

$A(1+x)^2 + B(1+8x)(1+x) + C(1+8x) = 1$

Let $x = -1$

$1 - 8 = -7$

$-7C = 1$

$C = -\frac{1}{7}$

$\frac{64}{49(1+8x)} - \frac{8}{49(x+1)} - \frac{1}{7(x+1)^2}$

$|8x| < 1$

$|x| < \frac{1}{8}$

Let $x = -\frac{1}{8}$

$\frac{49}{64}A = 1$

$A = \frac{64}{49}$

$z^2 + 7^2 = 4 + 49 = 54$

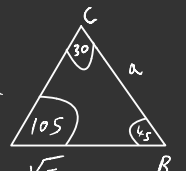
$2x^2 - 20x + 49$

$2(x^2 - 10x) + 49$

$2((x-5)^2 - 25) + 49$

$2(x-5)^2 - 50 + 49$

$2(x-5)^2 - 1$



$\frac{dy}{dx} = 0 \quad 4x - 20 = 0$

$4x = 20$

$x = 5$

$\frac{a}{\sin 105} = \frac{\sqrt{5}}{\sin 30}$

$\frac{\sqrt{5}}{\sin 30} \times \sin 105 = 4.31975\ldots$

$\frac{x+5}{(1-2x)^{\frac{1}{2}}} = (x+5)(1-2x)^{-\frac{1}{2}}$

$1 + (-\frac{1}{2})(-2x) + \frac{(-\frac{1}{2})(-\frac{3}{2})}{2}(-2x)^2 + \frac{(-\frac{1}{2})(-\frac{3}{2})(-\frac{5}{2})}{6}(-2x)^3$

$1 + x + \frac{3}{2}x^2 + \frac{5}{2}x^3$

$x + x^2 + \frac{3}{2}x^3$

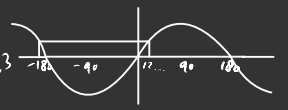
$5 + 5x + \frac{15}{2}x^2 + \frac{25}{2}x^3$

$\sin(x-15) = 0.22$

$-180 \leq x \leq 180 \quad -195 < x-15 < 165$

$x - 15 = 12.7090$

$x = 177.709$



$-180 - 12.\ldots = 177.7$

$-3(1+3x)^{-1}$

$1 + (-1)(3x) + \frac{(-1)(-2)}{2}(3x)^2$

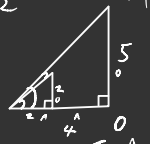
$1 - 3x + 9x^2$

$-3 + 9x - 27x^2$

$5 + 6x + \frac{17}{2}x^2 + 11x^3$

$|-2x| < 1$

$|x| < \frac{1}{2}$



$\frac{21x+2}{(1+3x)(1-2x)} = \frac{A(1+3x) + B(1-2x)}{}$

Let $x = -\frac{1}{3}$

$\frac{5}{3}B = -5$

$B = -3$

Let $x = 0.5$

$1 + 3(0.5)A = 21 \times 0.5 + 2$

$2.5A = 12.5$

$A = 5$

$\tan^{-1}(\frac{5}{4}) = 51$

$\tan A \quad \tan^{-1}(\frac{2}{2}) = 45$

$= 6.34019$

$1 + 2x + 4x^2$

$5 + 10x + 20x^2$

$5x - y - 20 = 0$

$y = 0 \quad 5x - 20 = 0$

$5x = 20 \quad (4,0)$

$x = 4$

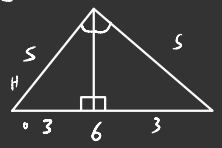
$x = 0 \quad -y - 20 = 0$

$y = -20 \quad (0,-20)$

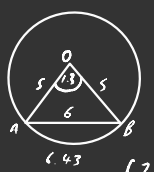
$\frac{4+0}{2} = 2 \quad \frac{0-20}{2} = -10$

$(2,-10)$

$\frac{1}{2}r^2\theta$



$\sin^{-1}\frac{3}{5} = 36.8\ldots \quad x2 = 73.739\ldots$



$16.0875\ldots$

$-12 = 4.0875\ldots$

$62.45228\ldots$

$-4.0875\ldots = 58.364\ldots$

$\frac{5}{1-2x} \quad \frac{-3}{1+3x}$

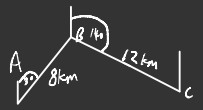
$-3 + 6x + 5 + 15x$

$21x + 2$

1

Calculate the distance in km of C from A if B is 8 km on a bearing of 030° from A and C is 12 km on a bearing of 140° from B. Give your answer to 1 decimal place. * (1 Point)

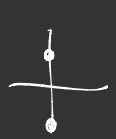
28/02



02/03/23

$y=f(x)$ 

$y=-f(x)$

$y=-2f(x)$ 

$P(0,4)$

$P(0,-4)$

$P(0,-8)$

$Q(4,1)$

$\left(1+\frac{x}{2}\right)^5 \quad \binom{5}{0}1^5+\binom{5}{1}1^4\times\left(\frac{x}{2}\right)+\binom{5}{2}1^3\times\left(\frac{x}{2}\right)^2+\binom{5}{3}1^2\times\left(\frac{x}{2}\right)^3$

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1

$1+\frac{5x}{2}+\frac{10x^2}{4}+\frac{10x^3}{8}+\dots$

$1+\frac{5}{2}x+\frac{5}{2}x^2+\frac{5}{4}x^3$

$1+\frac{x}{2}=1.2$

$x=0.4$

$\left(1+\frac{0.4}{2}\right)^5=1+\frac{5}{2}0.4+\frac{5}{2}0.4^2+\frac{5}{4}0.4^3$

$1+\frac{2}{2}+0.4+0.08$

$1+1+0.4+0.08$

2.48

$\tan\frac{\delta}{2}=\frac{\frac{1}{2}D}{d}$

$d\approx\frac{D}{\delta}$

If X is small and in radians

$\tan X\approx X$

$\cos x\approx 1-\frac{1}{2}x^2$

$\sin x\approx x$

$1-\frac{1}{2}x^2-x=x^2$

$x^2+x+\frac{1}{2}x^2-1=0$

$\frac{3}{2}x^2+x-1=0 \quad x=\frac{1}{2} \quad x=-2$

$5+\cos x-3\cos^2 x\approx a+\frac{b}{6}x^2$

$5+1-\frac{1}{2}x^2-3\left(1-\frac{1}{2}x^2\right)^2 \quad 1-x^2+\frac{1}{4}x^4$

$6-\frac{1}{2}x^2-3\left(1-x^2\right) \quad \cos 2\theta+\theta\sin 2\theta$

$6-\frac{1}{2}x^2-3+3x^2 \quad 1-\frac{(2\theta)^2}{2}+2\theta^2$

$3+\frac{5}{2}x^2 \quad 1-\frac{4\theta^2}{2}+2\theta^2$

$1-2\theta^2+2\theta^2=1$

$\left(1-\frac{1}{2}x^2\right)\left(1-\frac{1}{2}x^2\right) \quad 1-\frac{1}{2}x^2-\frac{1}{2}x^2+\frac{1}{4}x^4$

$y=(x+2)^2+3 \qquad f(x-a)=f(x)\binom{a}{0}$

$x^2+4x+4+3$

x^2+4x+7

$x^2+3 \qquad f(x)+b=f(x)\binom{0}{b}$

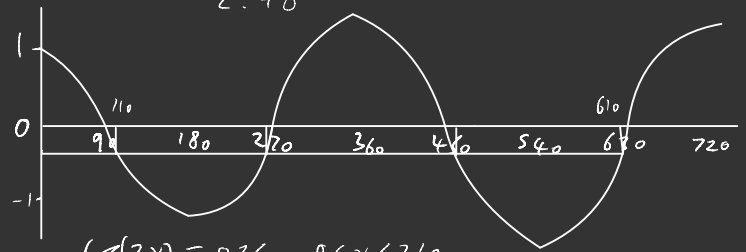
x^2+5

$y=x^2+3$

$af(x)$ = Vertical by a $\quad (x-2)^2$

$f(ax)$ = Horizontal by $\frac{1}{a}$

$2f(x)=2(x^2+3)=2x^2+6$



$\cos(2x)=-0.35 \quad 0<x<360$

$2x=110.487315 \quad 0<2x<720$

$x=55.243\dots$

$\cos(610)\approx-0.35 \quad 609.5126849\div 2=304.7563\dots$

$1^\circ=\frac{1^c}{60}=\frac{1}{1,000}$

$1\text{km}=1{,}000\text{m}=100{,}000\text{cm}$

$1\text{m}=1{,}000\text{cm}$

$180^\circ=\pi$

$\frac{\pi}{180}=1^\circ \quad 0.0033 \quad \approx\frac{1}{100} \quad 100.3346721$

$\frac{\frac{1}{60}\pi}{180}=0.000290888\,2087$

0.29089

$y = \frac{x}{2} + 3 - \frac{1}{x}$

$x = 2 \quad \frac{2}{2} + 3 - \frac{1}{2}$

$1 + 3 - \frac{1}{2} = 3\frac{1}{2} = \frac{7}{2}$

$\frac{1}{2}x - x^{-1}$

Sub 2 in

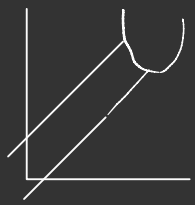
$2 - \frac{1}{2} = 3/2$

$y = 3/2\,x + c$

$7/2 = 3/2\,(2) + c$

$7/2 = 3 + c$

$c = 1/2$



$8\tan x - 3\cos x = 0$

$8\tan x = 3\cos x$

$\frac{8\sin x}{\cos} = 3\cos x$

$8\sin x = 3\cos^2 x$

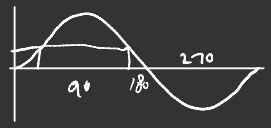
$3(1 - \sin^2 x) = 8\sin x$

$3 - 3\sin^2 x - 8\sin x = 0$

$3\sin^2 x + 8\sin x - 3 = 0$

$\sin x = 1/3$

$\sin x = -3$



$180 - 19.47 = 160.5$

Principal value

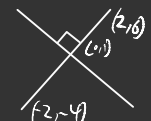
$y = \sin^{-1}(x) \quad -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

$y = \cos^{-1} x \quad 0 \leq y \leq \pi$

$\tan^{-1} x \quad -\frac{\pi}{2} < y < \frac{\pi}{2}$

$f(10, a)$ $(-2,-4)$ $(2,6)$

$m = (0,1)$



$\frac{6 - -4}{2 - -2} = \frac{10}{4} = \frac{5}{2}$

$y = \frac{5}{2}x + c$

$y = -\frac{2}{5}x + c$

$1 = -\frac{2}{5}(0) + c$

$c = 1$

$q = -\frac{2}{5}(-10) + 1$

$q = \frac{20}{5} + 1 = 5$



$\left|\tan\left(\frac{2}{9i}\right)\right| = 12.52$

$2(2^{2x}) - 17(2^x) + 8 = 0$

let $y = 2^x$

$2y^2 - 17y + 8 = 0$

$y = 8 \quad y = 1/2$

$2^x = 8 \quad 2^x = 1/2$

$x = 3 \quad x = -1$

$y = \frac{u}{v} \qquad u = yv$

$\frac{du}{dx} = y\frac{dv}{dx} + v\frac{dy}{dx}$

$\frac{du}{dx} - y\frac{dv}{dx} = v\frac{dy}{dx}$

$\frac{dy}{dx} = \frac{\frac{du}{dx} - y\frac{dv}{dx}}{v}$

$\frac{dy}{dx} = \frac{\frac{du}{dx} - \frac{u}{v}\frac{dv}{dx}}{v}$

$\times \frac{v}{v}$

$$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

quotient rule

$y = \frac{3x+1}{x+5} \qquad y = \frac{x^2}{7x-1} = \frac{u}{v}$

$u = 3x + 1$

$v = x + 5$

$\frac{du}{dx} = 3$

$\frac{dv}{dx} = 1$

$\frac{3(x+5) - (3x+1)}{(x+5)^2}$

$= \frac{14}{(x+5)^2}$

$\frac{2x(7x-1) - x^2(7)}{(7x-1)^2} = \frac{14x^2 - 2x - 7x^2}{49x^2 - 14x + 1}$

$\frac{7x^2 - 2x}{(7x-1)^2}$

$-x^2 + 2x + 8 = 0$

$x = 4 \quad x = -2$

$y = \frac{1 + x^2}{e^{5x}} = \frac{v}{}$

$\frac{e^{5x}(2x) - (1+x^2)(5e^{5x})}{(e^{5x})^2}$

$\frac{2xe^{5x} - 5e^{5x} - x^2 5e^{5x}}{(e^{5x})^2}$

$\frac{2x - 5 - 5x^2}{e^{5x}}$

$\frac{e^x(x-4)}{(x-3)(x-2)} = 0$

$e^x(x-4) = 0$

$e^x \neq 0 \quad x - 4 = 0$

$x = 4$

$$\left(2 - \tfrac{1}{2}x\right)^{12}$$

$$\binom{12}{0}2^{12} + \binom{12}{1}\left(2^{11}\right)\left(-\tfrac{1}{2}x\right) + \binom{12}{2}\left(2^{10}\right)\left(-\tfrac{1}{2}x\right)^2 +$$

$$\binom{12}{3}\left(2^{9}\right)\left(-\tfrac{1}{2}x\right)^3 + \binom{12}{4}\left(2^{9}\right)\left(-\tfrac{1}{2}x\right)^4$$

$$495 \times 512 \times \frac{1}{16}$$

$$\sim 15{,}840\,x^4$$

$-90 \le x \le 90$

$$x^{-1} + 4x^2$$

$$\frac{dy}{dx} = -1x^{-2} + 8x$$

$$-1x^{-2} + 8x = 0$$

$$8x = x^{-2}$$

$$8x - x^{-2} = 0$$

$$8x - \frac{1}{x^2} = 0$$

$$\frac{8x^3 - 1}{x^2} = 0$$

$$8x^3 - 1 = 0$$

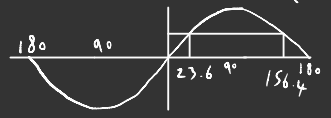
$$8x^3 = 1 \qquad x^3 = \tfrac{1}{8} = \tfrac{1}{2}$$

$$3\sin(2x) - 0.2 = 1$$

$$3\sin(2x) = 1.2$$

$$\sin(2x) = 0.4$$

$-180 \le 2x \le 180$



$$\frac{156.4}{2} = 78$$

$\sin 105 = \sin(60+45)$

$105 = 60 + 45$

$\sin 60 \cos 45 + \cos 60 \sin 45$

$$\frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2} + \frac{1}{2} \times \frac{\sqrt{2}}{2}$$

$$\frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4}$$

$$\frac{\sqrt{3}+1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{6}+\sqrt{2}}{2\times 2}$$

$$-\frac{\sqrt{2}}{2} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{-2}{2\sqrt{2}}$$

$$\frac{-1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{-\sqrt{2}}{2}$$

$$\frac{2+\sqrt{3}}{1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1} = \frac{2\sqrt{3}+3-\sqrt{3}-2}{\sqrt{3}-1} = \frac{\sqrt{3}+1}{\sqrt{3}-1}$$

$\sin(\theta - 30) = \cos\theta$

$\sin\theta\cos 30 - \cos\theta\sin 30$

$$\frac{\sqrt{3}}{2}\sin\theta - \tfrac{1}{2}\cos\theta = \cos\theta$$

$$\frac{\sqrt{3}}{2}\sin\theta = \frac{3}{2}\cos\theta$$

$$\sqrt{3}\sin\theta = 3\cos\theta$$

$$\sin\theta = \frac{3}{\sqrt{3}}\cos\theta$$

$$\tan\theta = \frac{3}{\sqrt{3}} = 60$$

$60 + 180 = 240$

$\cos\theta + 60 = \sin\theta$

$y = x^3 - 6x^2 + 3x + 10$

$\frac{dy}{dx} = 3x^2 - 12x + 3$

$x = 3$:

$3 \times 9 - 12 \times 3 + 3$

$= -6$

$y = -6x + c$

$3^3 - 54 + 9 + 10$

$27 - 54 + 9 + 10 = -8$

$-8 = -6(3) + c$

$-8 = -18 + c$

$-8 + 18 = c = 10$

$y = \frac{1}{6}x + c$

$-8 = \frac{3}{6} + c$

$c = -8 + \frac{1}{2} = -\frac{17}{2}$

$y = \frac{1}{6}x - \frac{17}{2}$

$6y = x - 51$

$6y - x + 51 = 0$

$x - 6y - 51 = 0$

$(3x-1)(x-1)$

$3x^2 - 3x - x + 1$

$3x^2 - 4x + 1 = 5$

$3x^2 - 4x - 4 = 0$

$2, -\frac{2}{3}$

$3^{2x-1} = 9^{-x}$

$3^{2x-1} = (3^2)^{-x}$

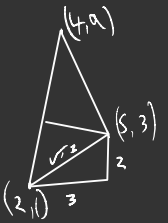
$3^{2x-1} = 3^{-2x}$

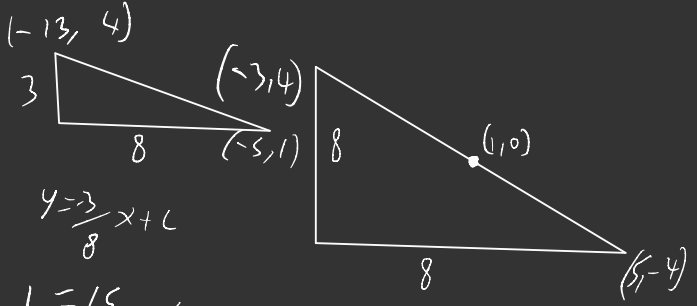
$2x - 1 = -2x$

$4x - 1 = 0$

$4x = 1$

$x = 1/4$





$y = \frac{-3}{8}x + c$

$1 = \frac{15}{8} + c$

$y = -7x + c$

$5 = -7(2) + c$

$6 = -14 + c$

$c = 20$

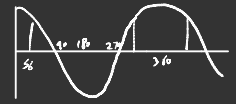
$(x-1)^2 + (y-2) = 25$

$4x - 3y + 2 = 0$

$\cos 3x = 0.545$

$0 \leq x \leq 360$

$3x = 56.975$



56

$360 - 56$

$360 + 56$

$720 - 56$

$720 + 56$

$1080 - 56 = \frac{1023.025}{3} = 341$

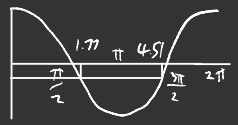


$\tan^{-1}\frac{3}{3} = 45$ +

$\tan^{-1}\left|\frac{-1}{4}\right| = 14.03$

$= 49$

$\frac{\sin x}{1 - \cos x} x$



$\int_1^2 \left[4 + 3x - x^3\right]$

$4x + \frac{3}{2}x^2 - \frac{1}{4}x^4$

$10\sin x - 3\cos x = 0 \quad -180 < x < 180$

$10\sin x = 3\cos x$

$10\tan x = 3$

$\tan x = \frac{3}{10}$

$\tan^{-1}\left(\frac{3}{10}\right) = 16.7$

$a\sin x + b\cos x = R\cos(\theta - \alpha)$

$a = R\sin\alpha$
$b = R\cos\alpha$
$R = \sqrt{a^2+b^2}$
$\tan\alpha = \frac{a}{b}$

$R\sin(x+\alpha)$
$a = R\cos\alpha$
$b = R\sin\alpha$
$R = \sqrt{a^2+b^2}$
$\tan\alpha = \frac{b}{a}$

$4\sin x + 5\cos x = R\cos(x-\alpha)$

$y = R\cos x\cos\alpha + R\sin x\sin\alpha$

$4\sin x = R\sin x\sin\alpha$

$4 = R\sin\alpha$
$5 = R\cos\alpha$
$\tan^{-1}\left(\frac{4}{5}\right) = 38.65\ldots$

$6\sin x + \cos x$

$6 = R\cos\alpha$
$1 = R\sin\alpha$

$\frac{\sin\alpha}{\cos\alpha} = \frac{1}{6}$

$\alpha = 9.46\ldots$

$\sqrt{1^2+6^2} = \sqrt{37}$

$\cos(x+\alpha) = \cos\alpha\cos a - \sin\alpha\sin a$
$6\sin x - 5\cos x = R\sin(x-\alpha)$
$\sin(x-\alpha) = \sin x\cos\alpha - \cos x\sin\alpha$

$6 = \cos\alpha$
$5 = \sin\alpha$
$\tan^{-1}\left(\frac{5}{6}\right) = 39.8$
$\sqrt{61}\sin(x-39.8)$

$3\sin x + 9\cos x$
$R\sin(x+\alpha)$

$R\sin x\cos\alpha + R\cos x\sin\alpha$

$3\sin x = R\cos x\cos\alpha$
$9\cos x = R\cos x\sin x$

$\frac{R\sin x}{R\cos x} = \tan^{-1}\left(\frac{9}{3}\right) = 71.565\ldots$

$3^2+9^2 = 90$
$\sqrt{90}\sin(x+72)$

$y = 6x + 2$

$y = x^3 - 3x^2 + 6x + 2$

$6x + k = x^3 - 3x^2 + 6x + 2$

$k = x^3 - 3x^2 + 2$

$3x^2 - 6x + 6 = 0$ $(-2, 8)$ $(-8, -3)$

$\frac{-3-5}{-8-(-2)} = \frac{-8}{-6} = \frac{4}{3}$

$y = -3x + c$

$5 = 6 + c$

$c = -1$

$y = \frac{4}{3}x + c$

$5 = \frac{4}{3}(-2) + c$

$c = 5 + \frac{8}{3} + c$

$5 = -\frac{8}{3} + c$

$\frac{x^2}{2} - x + c = y$

$2 = \frac{3^2}{2} - 3 + c$

$2 = \frac{9}{2} - 3 + c$

$c = 2 + 3 - \frac{9}{2} = \frac{1}{2}$

$y = \frac{1}{2}x^2 - x + \frac{1}{2}$

$y = \frac{25}{2} - 5 + \frac{1}{2}$

$\frac{25}{2} = 13$ $13 - 5 = 8$

$(-5, 1)$

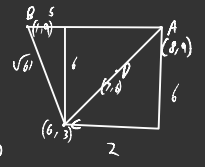
$1 = -\frac{3}{4}(5) + c$

$1 = -\frac{15}{4} + c$



12

$12^2 + 1^2 = 145$



$y = 3x + c$

$y = -\frac{1}{3}x + c$

$6 = -\frac{7}{3} + c$

$6 + \frac{7}{3} = c$

$y = -\frac{1}{3}x + \frac{25}{3}$

$3y = -x + 25$

$x + 3y - 25 = 0$

$10y = 5x + 16$

when $x = 0$ $(0, 1.6)$

$10y = 16$

$y = 1.6$

when $y = 0$

$5x + 16 = 0$ $(-3.2, 0)$

$5x = -16$

$x = -\frac{16}{5} = -3.2$



$3.2 \times 0.8 = 2.56$

$32 \times 08 = 2.56$

$|5i - mj| = |2mi + 2j|$

$5^2 + m^2 = (2m)^2 + 2^2$

$25 + m^2 = 4m^2 + 4$

$3m^2 = 21$

$m^2 = 7$

$m = \sqrt{7}$

$\sin x = \frac{-15}{17}$

$\sin^{-1}\left(\frac{-15}{17}\right)$

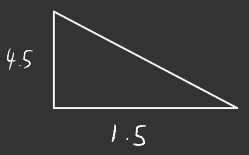
cos Ans=

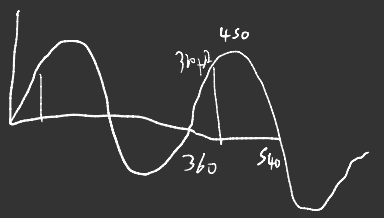
$(1+cx)^n = 1 + 24x + 264x^2$

$\binom{n}{0}1^n + \binom{n}{1}1^{n-1}cx + \binom{n}{2}1^{n-2}c^2x^2$

$n = n$ $\quad n^{n-1}C\,x$ $\quad \binom{n}{2}1^{n-2}c^2x^2$

$24 = n^nC$ $\quad \binom{n}{2}1^{n-2}c^2 = 264$





$\left(\frac{9}{2}\right)^2 + \left(\frac{3}{2}\right)^2 = \frac{81}{4} + \frac{9}{4} = \frac{90}{4}$

$2\log_3 x + 1 = \log_3(7x+6)$

$1 = 2\log_3\left(\frac{3x+6}{x}\right)$

$\frac{\sqrt{90}}{2} = \frac{\sqrt{9}\times\sqrt{10}}{2} = \frac{3\sqrt{10}}{2}$

$9x^2 + 4x^{-2}$

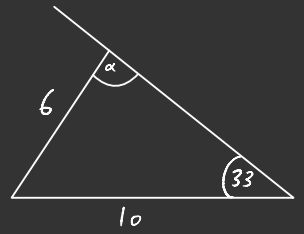
$-4x^{-1}$

$4x^{-2}$

$x = -1$

$9(1) + 4(-1^{-2})$ $\quad \frac{1}{-1^2} = \frac{1}{1} = 1$

$9 + 4 = 13$



$\frac{\sin(33)}{6} = \frac{\sin\alpha}{10}$

$\frac{\sin 33}{6} \times 10 = 0.9077$

$\sin^{-1}(\text{ANS}) = 65.2$



$6x + 4 = 100$

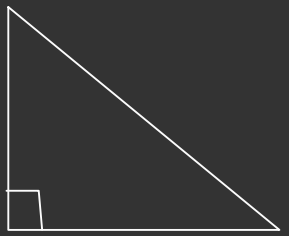
$6x = 96$

$x = \frac{96}{6} = 16$

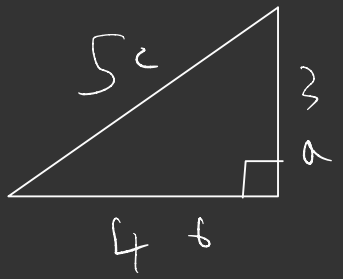
$a^2 = b^2 + c^2 - 2bc\cos A$

$(x-2)^2 = 10^2 + x^2 - 2\times 10x \times \cos(60)$

$x^2 - 4x + 4 = 100 + x^2 - 10x$



k (3,4,9)



$3 \times 3 = 9$

$4 \times 4 = 16$

$a^2 + b^2 = c^2$

$3^2 + 4^2 = 5^2$

$25 = c^2$

$c = 5$



Topics

- Enthalpy changes
- Measuring enthalpy changes
- Bond enthalpies
- Hess' law + enthalpy cycles
- Reaction rates
- Catalysts
- Boltzmann distribution
- Dynamic equilibrium and le Chatelier's principle
- equilibrium constant K_c

percentage yield
atom economy
percentage uncertainty